

[Total No. of Questions - 9] [Total No. of Printed Pages - 3]
Dec-25-0650

MCA-7305(iv) (Machine Learning)

MCA-3rd NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

SECTION-A

1. (a) Discuss the evolution of Machine learning. How can we differentiate traditional programming from machine learning? (6)
(b) What is feature reduction? List and explain two feature reduction techniques in brief. (6)
2. (a) Explain the concept of a hypothesis space in machine learning and discuss how inductive bias influences the choice of the most suitable hypothesis. (6)
(b) What are the different evaluation measures in machine learning? Why is cross-validation so popular? (6)

SECTION-B

3. a) What is Bayes theorem? Write important features in Bayesian learning methods. (6)
b) What is curse of dimensionality? How does it affect the performance of learning algorithms as the number of features increases? (6)
4. a) What is overfitting in machine learning, and how does careful model selection help prevent it? (6)

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- b) Explain the bias-variance trade-off in detail. (6)

SECTION-C

5. a) What is unsupervised learning? Explain any unsupervised learning technique used for clustering. (6)
b) What is vector quantization in the context of clustering, and how is it used to reduce the dimensionality? (6)
6. a) Explain dimensionality reduction and differentiate between linear and nonlinear methods with examples. (6)
b) What is PCA? Write and explain the steps for PCA. (6)

SECTION-D

7. a) A factory has two machines, M1 and M2 that produce items. Machine M1 produces 40% of the total output and machine M2 produces 60%. The probability that an item produced by M1 is defective is 0.01, and the probability that an item produced by M2 is defective is 0.03.
 - (i) Using the law of total probability, calculate the probability that a randomly chosen item from the factory is defective. (3)
 - (ii) If a randomly chosen item is found to be defective, what is the probability that it was produced by M1? Use Bayes' Rule and show your steps. (3)
- b) What is generative training in Machine Learning? Explain how it differs from discriminative training with suitable examples. (6)
8. Write short notes: (i) Logistic regression (ii) Bayesian network (2×6=12)

[P.T.O.]

SECTION-E (Compulsory)

9. a) Define Machine learning
- b) What is gradient descent?
- c) What is feature selection?
- d) Differentiate between training and testing.
- e) Can K-Means be applied directly to categorical data? Why or why not?
- f) State the main assumption underlying the Naive Bayes classifier.
- (6×2=12)